Q3 [6M]. Consider the following functions **HEAPIFY(), BUILD_HEAP()**, and **HEAP_SORT()**. Assuming that array index starts from 0 (zero), there are certain error(s) in some (or all) these functions. Identify and write the statements having error(s), and the corresponding correct statement(s).

```
Function HEAPIFY (arr, n, i)
    largest ← i
    left ← 2 * i
    right ← 2 * i + 1
    If (left < n AND arr[left] > arr[largest])
        largest ← left
    If (right < n AND arr[right] > arr[largest])
        largest ← right
    If largest ≠ i:
        SWAP(arr[i], arr[largest])
        HEAPIFY (arr, n, largest)
```

```
Function BUILD_HEAP (arr, n)
    For i = 0 to n/2
        HEAPIFY (arr, n, i)
```

```
Function HEAP_SORT (arr, n)
    BUILD_HEAP (arr, n)
    For i = 1 to n
        SWAP(arr[0], arr[i])
        n = n - 1
        HEAPIFY (arr, n, 0)
```

Ans 3:

Function Name	Statement having error	Correct statement	
HEAPIFY	left ← 2 * i	left ← 2 * i + 1	3M for all three statements
HEAPIFY	right ← 2 * i + 1	right ← 2 * i + 2	
HEAPIFY (set B only)	If (left < n AND arr[left] > arr[right])	If (left < n AND arr[left] > arr[largest])	
BUILD_HEAP	For i = 0 to n/2	For i = n/2-1 to 0	1M
HEAP_SORT	For i = 1 to n	For i = n-1 to 1	2M for both
HEAP_SORT	HEAPIFY (arr, n, 0)	HEAPIFY (arr, i, 0)	